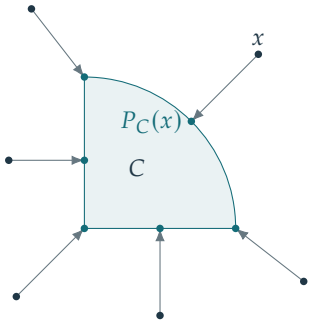
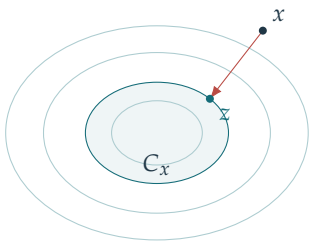
Projection onto a fixed set



$$x - P_C(x) \in N_C(P_C(x))$$

A sublevel set determined by the input

$$f(u,v) = \tfrac{1}{2}(u^2 + 2v^2)$$



$$z = \operatorname{prox}_f(x), \quad C_x = \{u : f(u) \le f(z)\}$$

$$z = P_{C_x}(x)$$